

SUBHAJIT MAHAPATRA

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github.com/Subhajit7710 | [LeetCode](#) | [Portfolio](#)

SUMMARY

Software Engineer and final-year MCA candidate with hands-on experience across the full software development lifecycle: analysis, design, coding, testing, debugging, documentation and cloud deployment. Skilled in Python, Java, SQL and JavaScript, using APIs and frameworks to build scalable AI solutions, integrating AI and LLMs into workflows, and analyzing data to generate insights. Collaborates effectively in cross-functional Agile (Scrum, Kanban) teams using Git/GitHub, CI/CD and AWS to build reliable technology that improves healthcare.

EDUCATION

Master of Computer Applications (MCA), Vellore Institute of Technology (VIT) | CGPA: 8.12 2025 – 2027
Bachelor of Computer Applications (BCA), Institute of Engineering & Management (IEM) | CGPA: 7.88 2021 – 2024

TECHNICAL SKILLS

Languages: Python, Java, C, SQL, JavaScript (ES6+), HTML5, CSS3

AI / ML: Artificial Intelligence (AI), Machine Learning (ML), Large Language Models (LLMs), NLP, RAG

ML Frameworks: PyTorch, TensorFlow, Hugging Face, PEFT (LoRA, QLoRA), Scikit-learn, FAISS, Pandas, NumPy

Web & Databases: Node.js, Express.js, React.js, REST APIs, Microservices, MySQL, PostgreSQL, MongoDB, Redis

Cloud & DevOps: AWS (EC2, S3, Lambda), Docker, Nginx, Linux, CI/CD, Continuous Delivery, Jenkins, Git, GitHub

AI Practices: Prompt Engineering & Optimization, AI Workflow Integration, AI-Driven Data Analysis, Responsible AI

Tools: GitHub Copilot, ChatGPT, VS Code (IDE), Jupyter Notebook, Postman, Bull Board

Core Concepts: DSA, OOP, DBMS, OS, System Design, SDLC, Agile (Scrum, Kanban), Testing, Debugging

EXPERIENCE

Frontend Developer Intern, Techplement

Jun 2024 – Jul 2024

- Increased checkout success rate by **75%**, measured as completed versus initiated payments, by integrating the Stripe Payment Gateway API with backend RESTful APIs in a React.js checkout flow.
- Improved UI responsiveness of the production website on desktop and mobile by cutting unnecessary re-renders with memoization (React.memo, useMemo, useCallback) and React Hooks in reusable, cross-browser components.

PROJECTS

FamilyCare: Real-Time Healthcare Medication Adherence Platform | [GitHub](#) | [Live](#)

Node.js, Express.js, React.js, MySQL, Redis, BullMQ, Socket.IO, JWT, Docker, Nginx, AWS EC2, OpenFDA API

- Built a scalable healthcare backend, as measured by **4 microservices serving 20 RESTful APIs**, by designing an API Gateway with health checks, HTTP 503 fallback and JWT + bcrypt authentication on MySQL and Sequelize ORM.
- Delivered fault-tolerant, real-time medication reminders, as measured by **6 live event types** and **3-attempt retries** with exponential backoff, by building a BullMQ job pipeline on Redis with JWT-authenticated Socket.IO and Redis Pub/Sub.
- Eliminated repeat third-party drug-data requests, as measured by a **24-hour** cache on every OpenFDA lookup, by integrating the OpenFDA API behind a Redis cache-aside layer in the health microservice.
- Simplified secure cloud deployment, as measured by a **one-command** setup exposing only **3 public ports**, by scripting AWS EC2 provisioning with Docker Compose, PM2, Nginx reverse proxy and HTTPS.

JustAdvisor AI: LLM-Powered Legal Reasoning & Verdict System | [GitHub](#)

Python, PyTorch, Hugging Face, PEFT, QLoRA, Mistral-7B, Legal-BERT, FAISS, RAG, NLTK

- Fine-tuned an LLM-based AI judge, as measured by token accuracy rising from **69.4% to 97%** and **90%** lower training loss, by applying QLoRA (4-bit NF4, LoRA r=16, alpha=32) to Mistral-7B, training only **0.09%** of parameters.
- Automated legal argument classification into **5 roles**, as measured by a **92%** drop in validation loss (0.72 to 0.055), by cleaning and analyzing a **900**-sample class-balanced dataset and fine-tuning Legal-BERT with FP16 mixed precision.
- Integrated AI into a legal-analysis workflow, as measured by grounded retrieval across **45 statutes and precedents**, by building a RAG pipeline with Hugging Face APIs, FAISS vector search and optimized, responsible-AI prompts.

CERTIFICATIONS & ACHIEVEMENTS

- Accenture Software Engineering Job Simulation (Forage), Jul 2024.
- Led and collaborated with a **4-member team** in a technical problem-solving round, coordinating strategy.